

Evaluation Report

Legal AI Assistant — Golden Set Evaluation · 18 questions · Run: 2026-07-17 10:18:09 UTC

■ IMPORTANT — READ BEFORE INTERPRETING THESE NUMBERS

*This evaluation run used a **stand-in LLM** for answer generation, not a real Claude/GPT/Gemini call. The development sandbox this project was built in has no configured **ANTHROPIC_API_KEY** (a credentials gap, not a network gap — `api.anthropic.com` is reachable from that sandbox). To produce a genuine, non-fabricated evaluation run without a real key, the LLM step was replaced with a simple word-overlap heuristic that selects and echoes the most relevant retrieved passage. Retrieval-side metrics below (Retrieval Recall, MRR, nDCG) are fully real, since the entire retrieval pipeline — parsing, chunking, hybrid search, MMR, re-ranking — is genuine production code with no stand-ins. Answer-side metrics (Answer Correctness, Faithfulness, Hallucination Rate) are real measurements of the stand-in LLM's crude behavior, not of Claude/GPT/Gemini's actual answer quality, and should not be read as a verdict on the system's real-world answer quality. Re-run `python -m evaluation.cli` with a real **ANTHROPIC_API_KEY** to get numbers that reflect actual answer quality.*

Methodology

18 questions (the **Golden Set**, see `Golden_Set.csv`) were hand-authored from the actual content of the sample legal document (a judgment concerning a Section 80G tax deduction dispute), covering direct fact lookups, statutory requirement questions, holdings, multi-page synthesis questions, and two deliberate “no-evidence-check” questions asking about facts genuinely absent from the document (to test the system's anti-hallucination refusal behavior). Each question was run through the complete pipeline: hybrid search (vector + BM25) → Reciprocal Rank Fusion → MMR diversification → re-ranking → confidence filtering → answer generation → citation formatting.

Aggregate Metrics

Metric	Value	What it measures
Retrieval Recall@5	1.0000	Fraction of expected passages found in top-5 retrieved chunks (retrieval quality, LLM-independent)
MRR	0.9167	Mean Reciprocal Rank of the first relevant retrieved chunk
nDCG@5	0.9420	Ranking quality of retrieved chunks (log-discounted relevance)
Citation Accuracy (Precision)	0.6667	Fraction of the answer's citations that match a real expected passage
Citation Recall	0.7407	Fraction of expected passages the answer actually cited
Answer Correctness	0.2576	Token-F1 vs. ground truth (SQuAD-style); binary for no-evidence-check questions
Faithfulness	0.1593	Fraction of answer sentences carrying a citation tag
Hallucination Rate	1.0000	Fraction of questions with an uncited/mismatched claim, or an incorrect refusal/non-refusal
Errored Questions	0 / 18	Questions where the pipeline itself threw an exception
Avg. Latency	0.0043s	Wall-clock time per question (retrieval + generation)

Interpreting These Numbers

Retrieval Recall@5 = 1.0, MRR = 0.92, nDCG@5 = 0.94 — these are genuinely strong and reflect the real, fully-implemented retrieval pipeline (hybrid vector+BM25 search, RRF fusion, MMR diversification, re-ranking). Every expected passage was findable within the top 5 retrieved chunks for every answerable question.

Citation Precision = 0.67, Citation Recall = 0.74 — moderate scores reflecting that the stand-in LLM sometimes cited zero passages (word-overlap heuristic occasionally scored no passage as relevant enough) rather than making a genuinely wrong citation — see the per-question table on the next page.

Answer Correctness = 0.26, Faithfulness = 0.16, Hallucination Rate = 1.0 — these are low almost entirely because the stand-in LLM does not reliably produce full-sentence, fully-cited answers the way a real instruction-following model does; it simply echoes the best-matching passage without careful citation formatting on every sentence. This is a measurement of the *stand-in's* weakness, not the system's prompt design (see Milestone 7's prompt engineering in the Approach Document) or its citation formatter (which is separately unit-tested and proven correct in the backend test suite, independent of which LLM produces the text).

Per-Question Results

ID	Category	Ans. Corr.	Cite P	Cite R	MRR	Faithful.	Halluc.	Latency (s)
q001	fact_lookup	0.27	1.00	1.00	1.00	0.20	1	0.005
q002	fact_lookup	0.08	0.00	0.00	0.50	0.20	1	0.004
q003	fact_lookup	0.08	0.00	0.00	0.50	0.20	1	0.004
q004	fact_lookup	0.17	1.00	1.00	1.00	0.20	1	0.004
q005	fact_lookup	0.00	0.00	0.00	1.00	0.00	1	0.004
q006	fact_lookup	0.42	1.00	1.00	1.00	0.20	1	0.004
q007	holding	0.38	1.00	1.00	1.00	0.20	1	0.004
q008	fact_lookup	0.29	1.00	1.00	1.00	0.17	1	0.004
q009	fact_lookup	0.24	1.00	1.00	1.00	0.17	1	0.004
q010	statutory_requirement	0.33	1.00	1.00	1.00	0.17	1	0.005
q011	fact_lookup	0.55	1.00	1.00	1.00	0.17	1	0.004
q012	statutory_requirement	0.61	1.00	1.00	1.00	0.20	1	0.004
q013	holding	0.27	1.00	1.00	1.00	0.20	1	0.004
q014	holding	0.41	1.00	1.00	1.00	0.20	1	0.004
q015	synthesis	0.20	0.00	0.00	0.50	0.20	1	0.004
q016	synthesis	0.34	1.00	0.33	1.00	0.20	1	0.004
q017	no_evidence_check	0.00	0.00	1.00	1.00	0.00	1	0.004
q018	no_evidence_check	0.00	0.00	1.00	1.00	0.00	1	0.004

Known Rigor Limitations (Stated Explicitly, Not Hidden)

- **Answer correctness** uses token-F1 (SQuAD-style) as a deterministic stand-in for an LLM-judge or semantic-similarity metric — it will under-score a correct answer phrased very differently from the ground truth.
- **Faithfulness** checks that each sentence carries a citation tag; it does NOT verify the cited passage actually entails the specific claim (that needs an LLM judge, as in RAGAS's real faithfulness metric).
- **Citation precision/recall** use page-range overlap matching against the Golden Set's expected citations, not exact equality.
- **This entire run used no real LLM API call** — see the callout at the top of this report.
- **Zero token usage reported** because the stand-in LLM doesn't report token counts; a real provider (Anthropic/OpenAI/Gemini) does, and this same report format would show real cost data with one.